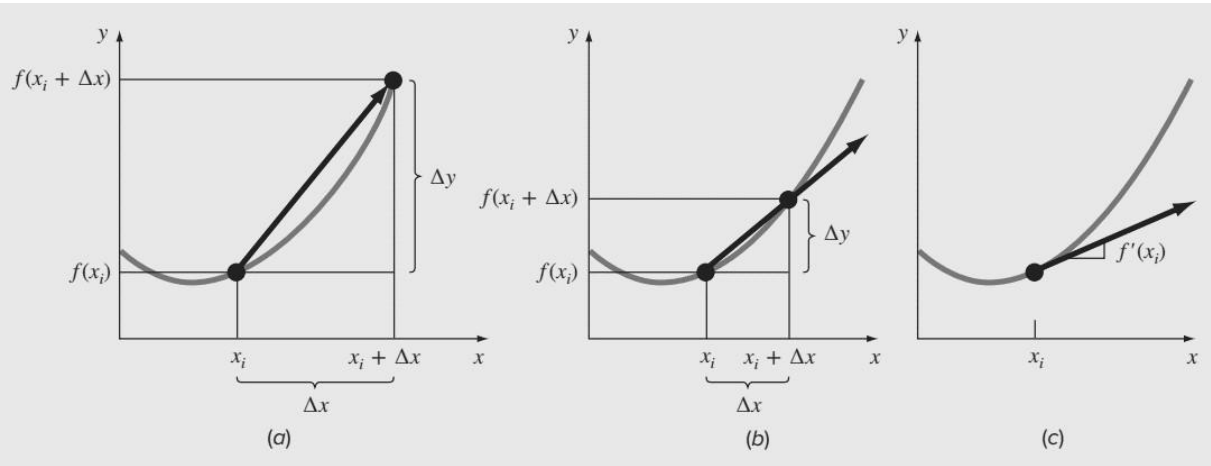
LAB 8: Numerical Differentiation

Objective: To numerically obtain the derivative of any function.

Theory: Calculus is the mathematics of change. Because engineers and scientists must continuously deal with systems and processes that change, calculus is an essential tool of our profession. Mathematically, the derivative, which serves as the fundamental vehicle for differentiation, represents the rate of change of a dependent variable with respect to an independent variable.



The graphical definition of a derivative: as Δx approaches zero in going from (*a*) to (*c*), the difference approximation becomes a derivative.

$$\frac{\Delta y}{\Delta x} = \frac{f(x_i + \Delta x) - f(x_i)}{\Delta x}$$

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{f(x_i + \Delta x) - f(x_i)}{\Delta x}$$

Law	Equation	Physical Area	Gradient	Flux	Proportionality
Fourier's law	$q = -k\frac{dT}{dx}$	Heat conduction	Temperature	Heat flux	Thermal Conductivity
Fick's law	$J = -D\frac{dc}{dx}$	Mass diffusion	Concentration	Mass flux	Diffusivity
Darcy's law	$q = -k\frac{dh}{dx}$	Flow through porous media	Head	Flow flux	Hydraulic Conductivity
Ohm's law	$J = -\sigma\frac{dV}{dx}$	Current flow	Voltage	Current flux	Electrical Conductivity
Newton's viscosity law	$\tau = \mu\frac{du}{dx}$	Fluids	Velocity	Shear Stress	Dynamic Viscosity
Hooke's law	$\sigma = E\frac{\Delta L}{L}$	Elasticity	Deformation	Stress	Young's Modulus

Methodology to find derivative numerically-

Using Taylor series expansion,

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \cdots$$

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} - \frac{f''(x_i)}{2!}h + O(h^2)$$

Also,

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} + O(h)$$

Similarly,

$$f''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{h^2} + O(h)$$

Table for different order of accuracy-

First Derivative	Error
$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h}$	$O(h)$
$f'(x_i) = \frac{-f(x_{i+2}) + 4f(x_{i+1}) - 3f(x_i)}{2h}$	$O(h^2)$
Second Derivative	
$f''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{h^2}$	$O(h)$
$f''(x_i) = \frac{-f(x_{i+3}) + 4f(x_{i+2}) - 5f(x_{i+1}) + 2f(x_i)}{h^2}$	$O(h^2)$
Third Derivative	
$f'''(x_i) = \frac{f(x_{i+3}) - 3f(x_{i+2}) + 3f(x_{i+1}) - f(x_i)}{h^3}$	$O(h)$
$f'''(x_i) = \frac{-3f(x_{i+4}) + 14f(x_{i+3}) - 24f(x_{i+2}) + 18f(x_{i+1}) - 5f(x_i)}{2h^3}$	$O(h^2)$
Fourth Derivative	
$f''''(x_i) = \frac{f(x_{i+4}) - 4f(x_{i+3}) + 6f(x_{i+2}) - 4f(x_{i+1}) + f(x_i)}{h^4}$	$O(h)$
$f''''(x_i) = \frac{-2f(x_{i+5}) + 11f(x_{i+4}) - 24f(x_{i+3}) + 26f(x_{i+2}) - 14f(x_{i+1}) + 3f(x_i)}{h^4}$	$O(h^2)$

Problem Statement.

Estimate the derivative of the below function for $O(h)$, $O(h^2)$ using forward differencing-

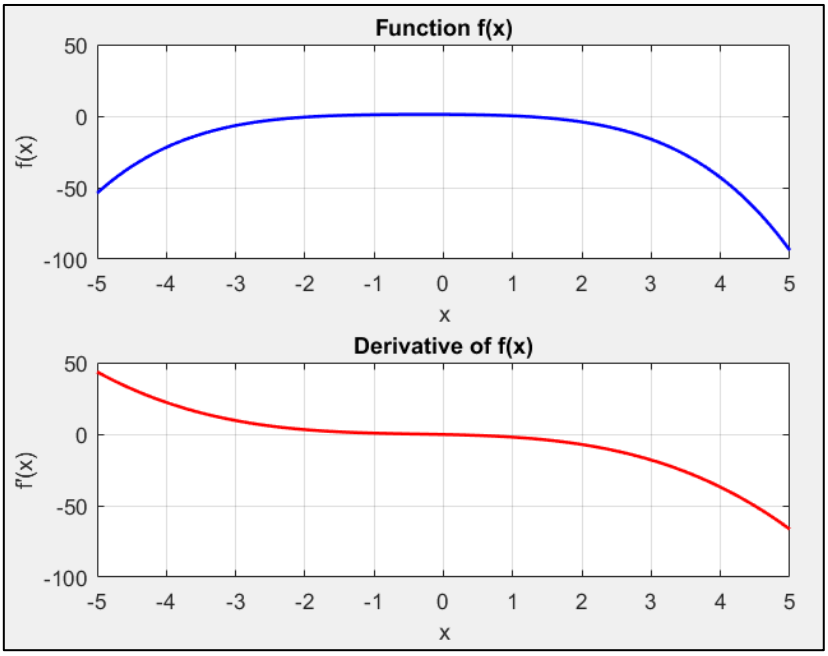
$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$$

at $x = 0.5$ using finite-differences and a step size of $h = 0.25$.

```
% Define the function
clear all; close all; clc;
f = @(x) -0.1*x.^4 - 0.15*x.^3 - 0.5*x.^2 - 0.25*x + 1.2;
x_values = linspace(-5, 5, 100);
y_values = f(x_values);
h = 0.25;

% forward difference method
df_approx = (f(x_values + h) - f(x_values)) / (h);

figure;
subplot(2,1,1);
plot(x_values, y_values, 'b', 'LineWidth', 1.5);
title('Function f(x)');
xlabel('x');
ylabel('f(x)');
grid on;
subplot(2,1,2);
plot(x_values, df_approx, 'r', 'LineWidth', 1.5);
title('Derivative of f(x)');
xlabel('x');
ylabel('f''(x)');
grid on;
```



Assignment -

One numerical as suggested by Faculty.

Sample numerical.

Develop an M-file to obtain first-derivative estimates for unequally spaced data. Test it with the following data:

x	0.6	1.5	1.6	2.5	3.5
$f(x)$	0.9036	0.3734	0.3261	0.08422	0.01596

where $f(x) = 5e^{-2x}\,x$. Compare your results with the true derivatives.

Code:

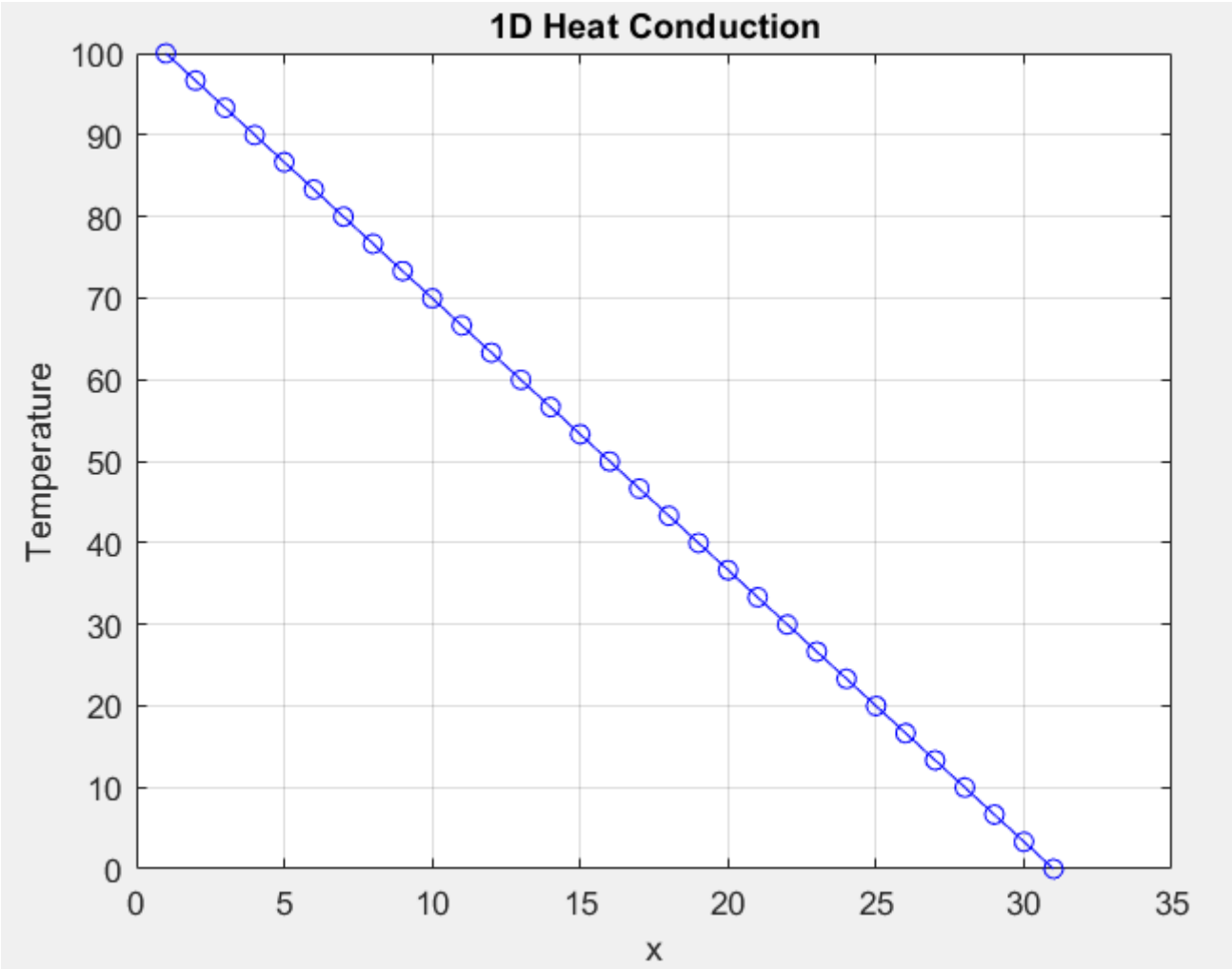
```
clc;
clear all;

x = [0.5 1.0 1.5 2.0 2.5 3.0 3.5];
y = [0.9198 0.6765 0.3735 0.1830 0.0843 0.0372 0.01596];

h = 0.5;

for i = 1:length(x)-1
   df(i) = (y(i+1) - y(i)) / h;
end

disp('  x     f(x)    f''(x)');
for i = 1:length(df)
   fprintf('%5.2f  %6.4f  %6.4f\n', x(i), y(i), df(i));
end
```

Result:

x	f(x)	f'(x)
0.50	0.9198	-0.4866
1.00	0.6765	-0.6060
1.50	0.3735	-0.3810
2.00	0.1830	-0.1974
2.50	0.0843	-0.0942
3.00	0.0372	-0.0425